4.5.2. Definition

"The accelerator coefficient is the ratio between induced investment and an initial change in consumption."

Assuming the expenditure of ₹50crores on consumption goods, if industries lead to an investment of ₹ 100 crores in investment goods industries, we can say that the accelerator is 2.

$$\text{Accelerator} = \frac{100}{50} = 2$$

4.5.3. Assumptions

1. Absence of excess capacity in consumer goods industries.
2. Constant capital - output ratio
3. Increase in demand is assumed to be permanent
4. Supply of funds and other inputs is quite elastic
5. Capital goods are perfectly divisible in any required size.

4.5.4. Operation of the Acceleration Principle

Let us consider a simple example. The operation of the accelerator may be illustrated as follows.

Let us suppose that in order to produce 1000 consumer goods, 100 machines are required. Also suppose that working life of a machine is 10 years. This means that every year 10 machines have to be replaced in order to maintain the constant flow of 1000 consumer goods. This might be called replacement demand.

Suppose that demand for consumer goods rises by 10 percent (ie from 1000 to 1100). This results in increase in demand for 10 more machines. So that total demand for machines is 20. (10 for replacement and 10 for meeting increased demand). It may be noted here a 10 percent increase in demand for consumer goods causes a 100 percent increase in demand for machines (from 10 to 20). So we can conclude even a mild change in demand for consumer goods will lead to wide change in investment.

Diagrammatic illustration:

Operation of Accelerator.

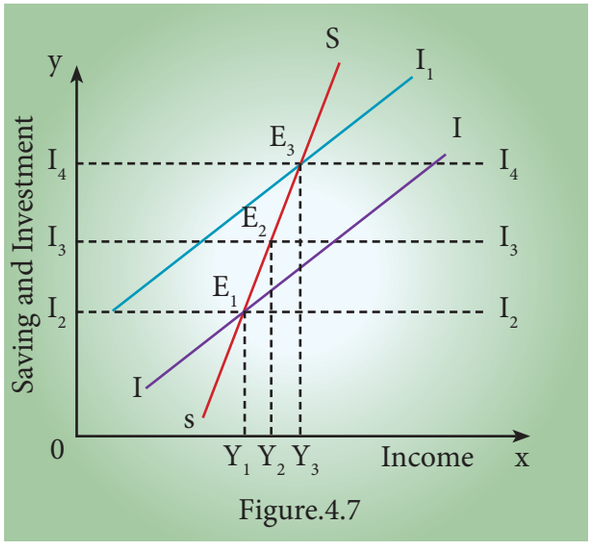


Figure.4.7

SS is the saving curve. II is the investment curve. At point E_1, the economy is in equilibrium with OY_1 income. Saving and investment are equal at OI_2. Now, investment is increased from OI_2 to OI_4. This increases income from OY_1 to OY_3, the equilibrium point being E_3. If the increase in investment by I_2 I_4 is purely exogenous, then the increase in

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